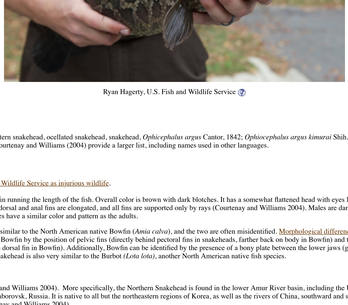
Channa argus
(Northern Snakehead)
Fishes
Exotic

Collection Info
Point Map
Species Profile
Animated Map
Impacts

Image 1 of 3



Ryan Hagerty, U.S. Fish and Wildlife Service

Channa argus (Cantor, 1842)

Common name: Northern Snakehead

Synonyms and Other Names: Amur snakehead, eastern snakehead, ocellated snakehead, snakehead, *Ophicephalus argus* Cantor, 1842; *Ophiocephalus argus amurensis* Shih, 1936; *Ophicephalus argus warpachowskii* Berg, 1909; *Ophicephalus pekinensis* Basilewsky, 1855. Courtenay and Williams (2004) provide a larger list, including names used in other languages.

Taxonomy: available through ITIS

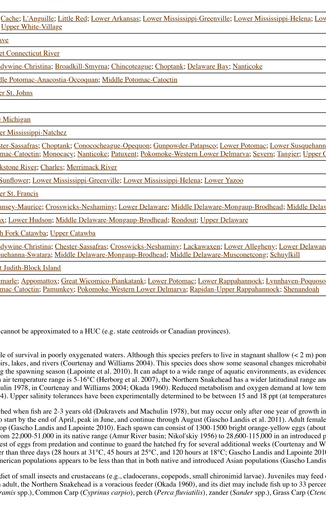
Injurious: This species is listed by the U.S. Fish and Wildlife Service as injurious wildlife.

Identification: A long, thin fish with a single dorsal fin running the length of the fish. Overall color is brown with dark blotches. It has a somewhat flattened head with eyes located in a dorsolateral position on the anterior part of the head; anterior nostrils are present and tubular; dorsal and anal fins are elongated, and all fins are supported only by rays (Courtenay and Williams 2004). Males are darker in color, and have a broader head, as compared to females (Gascho Landis and Lapointe 2010). Juveniles have a similar color and pattern as the adults.

Snakeheads (family Channidae) are morphologically similar to the North American native bowfin (*Amia calva*), and the two are often misidentified. Morphological differences used for identification between the two are depicted here. Snakeheads can be distinguished from the bowfin by the position of pelvic fins (directly behind pectoral fins in snakeheads, farther back on body in Bowfin) and the size of the anal fin (elongate and similar in size to dorsal fin in snakeheads, short and much smaller than dorsal fin in Bowfin). Additionally, Bowfin can be identified by the presence of a bony plate between the lower jaws (gular plate) and a distinctive method of swimming through undulation of the dorsal fin. The Northern Snakehead is also very similar to the Burbot (*Lota lota*), another North American native fish species.

Size: Maximum size exceeds 85 cm (33 inches).

Native Range: China, Russia and Korea (Courtenay and Williams 2004). More specifically, the Northern Snakehead is found in the lower Amur River basin, including the Ussuri River basin and Khanka Lake; the Sungari River in Manchuria; and, the Tungushka River at Khabarovsk, Russia. It is native to all but the northeastern regions of Korea, as well as the rivers of China, southward and southwestward to the upper tributaries of the Yangtze River basin in northeastern Yunnan Province (Courtenay and Williams 2004).



Hydrologic Unit Codes (HUCs) Explained
Interactive maps: Point Distribution Maps

Nonindigenous Occurrences:

Table 1. States with nonindigenous occurrences, the earliest and latest observations in each state, and the tally and names of HUCs with observations†. Names and dates are hyperlinked to their relevant specimen records. The list of references for all nonindigenous occurrences of *Channa argus* are found here.

State	First Observed	Last Observed	Total HUCs with observations†	HUCs with observations†
AR	2008	2025	11	Big; Cache; L'Anguille; Little Red; Lower Arkansas; Lower Mississippi Greenville; Lower Mississippi-Helena; Lower St. Francis; Lower White; Lower White-Bayou Des Arc; Upper White-Village
CA	1997	1997	1	Mojave
CT	2017	2017	1	Outlet Connecticut River
DE	2010	2025	6	Brandywine-Christina; Broadkill-Smyrna; Chincoteague; Choptank; Delaware Bay; Nanticoke
DC	2005	2024	2	Middle Potomac-Anacostia-Occoquan; Middle Potomac-Catoctin
FL	2000	2000	1	Upper St. Johns
GA	2019	2019	*	
IL	2004	2004	1	Lake Michigan
LA	2023	2023	1	Lower Mississippi-Natchez
MD	2002	2025	15	Chester-Sassafras; Choptank; Conococheague-Opequon; Gunpowder-Patapsco; Lower Potomac; Lower Susquehanna; Middle Potomac-Anacostia-Occoquan; Middle Potomac-Catoctin; Monocacy; Nanticoke; Patuxent; Pokomoke-Western Lower Delmarva; Severn; Tangier; Upper Chesapeake Bay
MA	2001	2023	3	Blackstone River; Charles; Merrimack River
MS	2017	2025	4	Big Sunflower; Lower Mississippi Greenville; Lower Mississippi-Helena; Lower Yazoo
MO	2019	2024	1	Lower St. Francis
NJ	2009	2024	5	Cohansey-Maurice; Crosswicks-Neshaminy; Lower Delaware; Middle Delaware-Mongaup-Brodhead; Middle Delaware-Musconetcong
NY	2005	2021	5	Bronx; Lower Hudson; Middle Delaware-Mongaup-Brodhead; Rondout; Upper Delaware
NC	2002	2007	2	South Fork Catawba; Upper Catawba
PA	1880	2025	12	Brandywine-Christina; Chester-Sassafras; Crosswicks-Neshaminy; Lackawaxen; Lower Allegheny; Lower Delaware; Lower Monongahela; Lower Susquehanna; Lower Susquehanna-Swatara; Middle Delaware-Mongaup-Brodhead; Middle Delaware-Musconetcong; Schuylkill
RI	2024	2024	1	Pesut Judith-Block Island
VA	2004	2025	12	Albemarle; Appomattox; Great Wicomico-Piankatank; Lower Potomac; Lower Rappahannock; Lynnhaven-Poquoson; Middle Potomac-Anacostia-Occoquan; Middle Potomac-Catoctin; Pamunkey; Pokomoke-Western Lower Delmarva; Rapidan-Upper Rappahannock; Shenandoah

Table last updated 12/3/2025

† Populations may not be currently present.

* HUCs are not listed for states where the observation(s) cannot be approximated to a HUC (e.g. state centroids or Canadian provinces).

Ecology: *Channa argus* is an obligate air-breather, capable of survival in poorly oxygenated waters. Although this species prefers to live in stagnant shallow (< 2 m) pools or swamps with mud substrate or aquatic vegetation and slow muddy streams, it also occurs in canals, reservoirs, lakes, and rivers (Courtenay and Williams 2004). This species does show some seasonal changes microhabitat selection and preference, utilizing deeper water in winter months and shallow areas with macrophytes during the spawning season (Lapointe et al. 2010). It can adapt to a wide range of aquatic environments, as evidenced by the spread of reproducing, introduced populations throughout Asia and Japan. While its optimum maximum air temperature range is 5-16°C (Herborg et al. 2007), the Northern Snakehead has a wider latitudinal range and temperature tolerance (0 to >30°C, including frost days) than other snakehead species (Dukravets and Machulin 1978, in Courtenay and Williams 2004; Okada 1960). Reduced metabolism and oxygen demand at low temperatures allows this species to survive extended periods of ice cover (Frank 1970, in Courtenay and Williams 2004). Upper salinity tolerances have been experimentally determined to be between 15 and 18 ppt (at temperatures of 15-24°C; Angelone et al. 2015).

In its native range, reproductive maturity is typically reached when fish are 2-3 years old (Dukravets and Machulin 1978), but may occur only after one year of growth in some introduced populations (USACE 2011). In the U.S., Northern Snakehead spawning has been observed to start by the end of April, peak in June, and continue through August (Gascho Landis et al. 2011). Adult females build circular floating nests from clipped aquatic plants and release their pelagic, nonadhesive, buoyant eggs on top (Gascho Landis and Lapointe 2010). Each spawn can consist of 1300-1500 bright orange-yellow eggs (about 1.8 mm diameter), with up to five spawns occurring within a year. Northern Snakehead fecundity can range from 22,000-51,000 in its native range (Amur River basin; Nikol'skiy 1956) to 28,600-115,000 in an introduced population (Syr Dar'ya basin, Turkmenistan/Uzbekistan; Dukravets and Machulin 1978). Both parents guard the nest of eggs from predation and continue to guard the hatched fry for several additional weeks (Courtenay and Williams 2004, Gascho Landis and Lapointe 2010). Depending on water temperature, eggs may hatch in fewer than three days (28 hours at 31°C, 45 hours at 25°C, and 120 hours at 18°C; Gascho Landis and Lapointe 2010). Larvae experience rapid growth after their first two weeks, though overall individual growth rate in North American populations appears to be less than that in both native and introduced Asian populations (Gascho Landis et al. 2011).

Fry initially feed on zooplankton, before moving on to a diet of small insects and crustaceans (e.g., cladocerans, copepods, small chironimid larvae). Juveniles may feed on small fish, including Goldfish (*Carassius* spp.) and roach (*Rutilis* spp.; Courtenay and Williams 2004). As an adult, the Northern Snakehead is a voracious feeder (Okada 1960), and its diet may include fish up to 33 percent of its body length (Courtenay and Williams 2004). Adult prey items include loach (*Cobitis* spp.), bream (*Abramis* spp.), Common Carp (*Cyprinus carpio*), perch (*Perca fluviatilis*), zander (*Sander* spp.), Grass Carp (*Ctenopharyngodon idella*), various catfishes, cray fish, dragonfly larvae, beetles, and frogs.

Although the Northern Snakehead can survive up to four days out of the water, overland migration is only possible for juveniles (Courtenay and Williams 2004). The rounded body of the adult Northern Snakehead is not as conducive to overland migration as observed in more horizontally flattened snakehead species.

Means of Introduction: Potential pathway of introduction: Unauthorized intentional release from aquariums or live food markets

Northern Snakehead likely arrived in U.S. waters by importation for the live food fish market (Angelone et al. 2015). Unauthorized intentional release from this trade, as was the case in the founding individuals of the Crofton pond population in Maryland, continues to be the major mechanism for introduction (Courtenay and Williams 2004). The Northern Snakehead has become widely popular in ethnic markets and restaurants over the last two decades, such that this species comprised the greatest volume and weight of all live snakehead species imported into the U.S. until 2001 (Courtenay and Williams 2004, Angelone et al. 2015). In Canada, Herberg et al. (2007) identified two watersheds in the Toronto area along Lake Ontario to be at the greatest risk for Northern Snakehead introduction from the live fish trade; the Rideau River watershed and Cedar Creek watershed (between Lake Erie and Lake St. Clair) posed additional vectors for introduction. Snakeheads' resilient nature reportedly makes them more desirable than carps for ceremonial release, and some interest in recreational fishing may also exist (Mendoza-Alfaro et al. 2009, Angelone et al. 2015).

Recognized as a highly injurious species, importation and cross-border transport of Northern Snakehead was prohibited in the U.S. by a 2002 listing under the Lacey Act and has been subsequently banned in Ontario. Nevertheless, cases of Northern Snakehead for sale in areas where possession is illegal are not uncommon (Angelone et al. 2015). Accidental release during transport of live fish is possible, but its probability is unknown (Mendoza-Alfaro et al. 2009).

Status: *Channa argus* is established in Delaware, Virginia, Maryland, Pennsylvania, New York, New Jersey, and Arkansas but is **not** established in California, Florida, Illinois, Massachusetts, and North Carolina where a few individual fish have been collected. However, the Northern Snakehead was eradicated from the Crofton pond in Maryland where it was ordinally established. The species is well established in the Potomac River and several of its tributaries in Virginia and Maryland (Starnes et al. 2011). Although young fish were found, the status of the Philadelphia population is uncertain. Officials believe fish may have gotten into the lower Schuylkill River and Delaware River in Pennsylvania and see no practical means to eradicate them. In March 2009, the population in Little Piney Creek drainage received an eradication attempt with the application of rotenone to more 700 km of creeks, ditches, and backwaters. However, more snakeheads have been found since this effort (L. Holt, pers.comm.). The population in Catlin Creek, New York was also treated with rotenone.

The Northern Snakehead's broad physiological tolerances, capacity to overwinter—including survival under ice, varied and flexible diet throughout at all life history stages, predatory and competitive nature, high fecundity, and parental investment in offspring, give this species a suite of favorable attributes for establishment once introduced. Northern Snakehead can adapt to a wide range of aquatic habitats and has been predicted to have high environmental suitability in northern U.S. and southern Canada, including abundant potential habitat in the Great Lakes (Herborg et al. 2007, Mendoza-Alfaro et al. 2009, Angelone et al. 2015).

Historical imports to the U.S. have come from a wide range of source populations, including Nigeria, Thailand, Indonesia, China, and Korea (Angelone et al. 2015). Orrell and Weigt (2005) found seven unique mitochondrial DNA haplotypes, none of which were shared, among the five U.S. populations they surveyed, indicating separate introduction events and source populations for each. Such high genetic diversity among introduced populations can promote their establishment and spread (Lee 2002, Sanders 2010).

Impact of Introduction:

Summary of species impacts derived from literature review. Click on an icon to find out more...

Specific impacts are unknown surrounding the Potomac population. These predatory fishes may compete with native species for food and habitat. Juveniles eat zooplankton, insect larvae, small crustaceans, and the fry of other Channa snakeheads almost exclusively on other fish (>97% of diet), with the remainder of their diet composed of crustaceans, frogs, small reptiles, and sometimes small birds and mammals (Courtenay and Williams 2004; Saylor et al 2012). Adult snakeheads show significant diet overlap with largemouth bass (*Micropterus salmoides*), with both consuming a large proportion of fundulids and other centrarchids in the lower Potomac River (Saylor et al. 2012).

Remarks: There is no evidence that juveniles or adult snakeheads escaped from the Crofton ponds. The Northern Snakehead has a wider latitudinal range and temperature tolerance than other snakehead species. It also seems to be adaptable to a wide range of aquatic environments, as evidenced by the spread of reproducing, introduced populations in Asia and Japan.

A specimen collected from Lake Wylie, North Carolina, in 2009 was originally identified as *Channa argus*, but later genetic work combined with a closer morphological analysis determined the specimen to be *Channa maculata* (NCSM 53258; W. Starnes, personal communication).

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Other Resources:
USGS Snakehead photo gallery

USFWS Injurious Wildlife

Snakeheads - Federal Register Final Rule

Do you know the difference (ID guide)

Snakeheads (Pisces, Channidae) - A Biological Synopsis and Risk Assessment by Courtenay and Williams (March 2004) or download a 7.6 MB PDF version.

Global Invasive Species Database Factsheet

US Fish and Wildlife Service Ecological Risk Screening Summary for *Channa argus*

FishBase Summary

Author: Fuller, P.L., Benson, A.J., Nunez, G., Fusaro, A., and Neilson, M.

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